In [ ]:
```
PROBLEM STATEMENT :- Classification using Deep neural network: Binary classification u
Classify movie reviews into positive" reviews and "negative" reviews, just based on th
Use IMDB dataset
```

In [1]:
```
import tensorflow as tf
from mlxtend.plotting import plot_confusion_matrix
from sklearn import metrics
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
from tqdm.notebook import tqdm
import warnings
warnings.filterwarnings("ignore")
```

In [2]:
```
vocab_size = 10000
max_len = 200
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.imdb.load_data(num_words=voca
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.
npz
17464789/17464789 [==============================] - 0s 0us/step
```

In [3]:
```
x_train.shape, y_train.shape, x_test.shape, y_test.shape
```

Out[3]:
```
((25000,), (25000,), (25000,), (25000,))
```

In [4]:
```
x_train = tf.keras.preprocessing.sequence.pad_sequences(x_train, maxlen=max_len)
x_test = tf.keras.preprocessing.sequence.pad_sequences(x_test, maxlen=max_len)
```

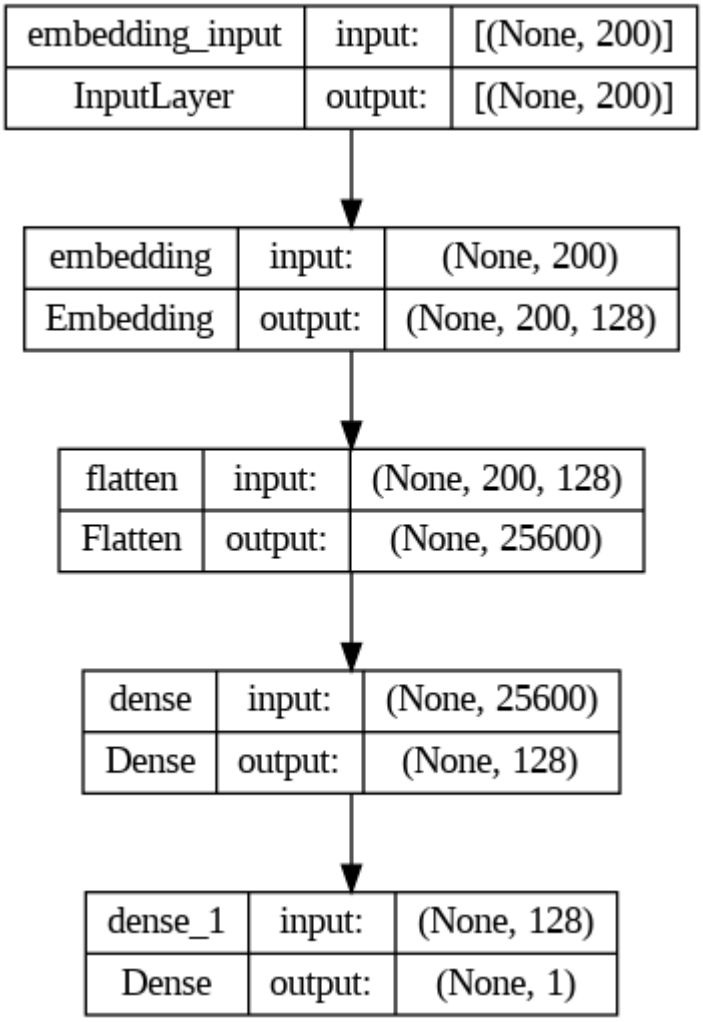
In [5]:
```
x_train.shape, y_train.shape, x_test.shape, y_test.shape
```

Out[5]:
```
((25000, 200), (25000,), (25000, 200), (25000,))
```

In [6]:
```
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(vocab_size, 128, input_length=max_len),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(1, activation='sigmoid')
])
```

In [7]:
```
tf.keras.utils.plot_model(model, show_shapes=True)
```

Out[7]:



In [8]:
```
model.summary()
```

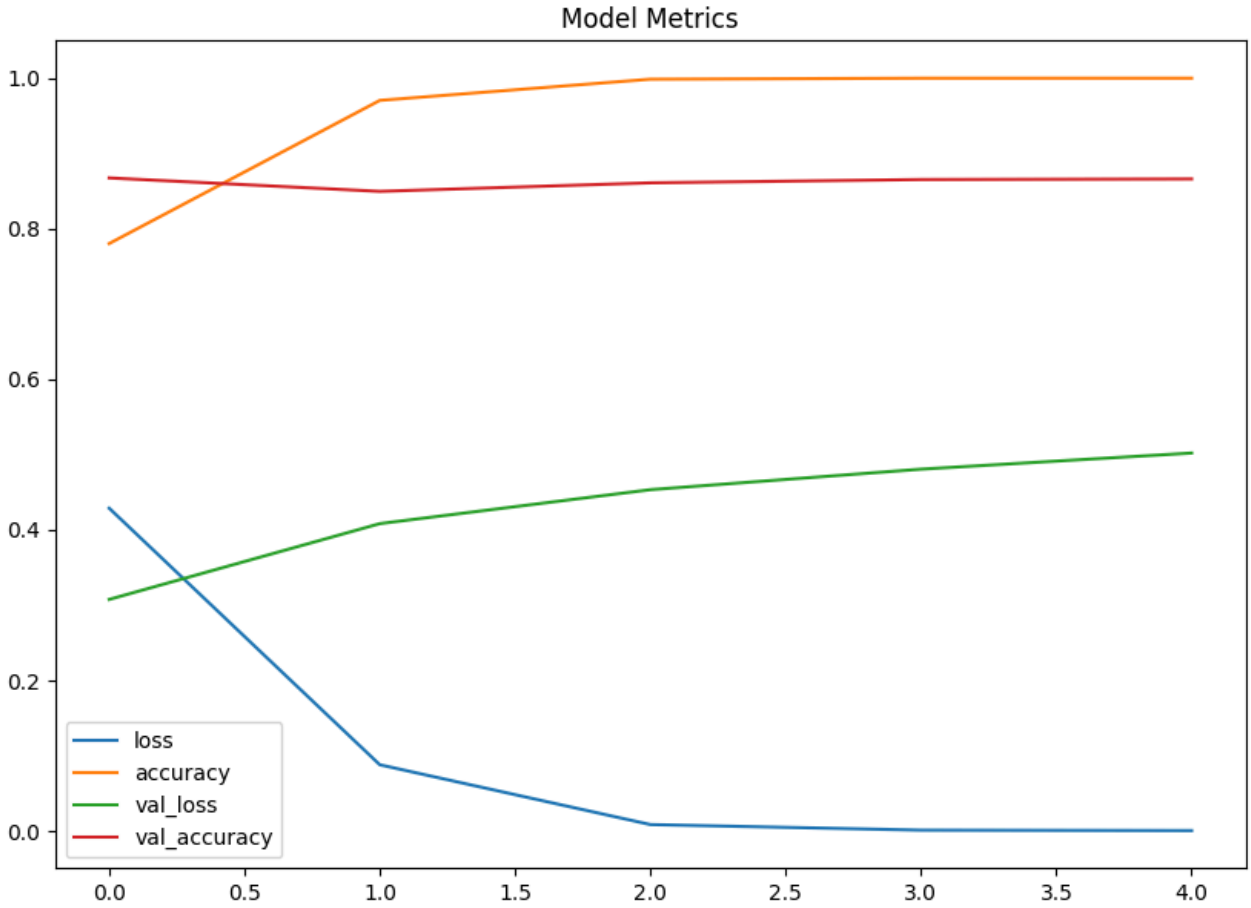
```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 embedding (Embedding)       (None, 200, 128)          1280000

 flatten (Flatten)           (None, 25600)             0

 dense (Dense)               (None, 128)               3276928

 dense_1 (Dense)             (None, 1)                 129

=================================================================
Total params: 4557057 (17.38 MB)
Trainable params: 4557057 (17.38 MB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
```

In [9]:
```
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

In [10]:
```
history = model.fit(x_train, y_train, batch_size=128, epochs=5, validation_data=(x_tes
```

```
Epoch 1/5
196/196 [==============================] - 18s 75ms/step - loss: 0.4287 - accuracy: 0.7
803 - val_loss: 0.3076 - val_accuracy: 0.8675
Epoch 2/5
196/196 [==============================] - 10s 50ms/step - loss: 0.0879 - accuracy: 0.9
705 - val_loss: 0.4082 - val_accuracy: 0.8496
Epoch 3/5
196/196 [==============================] - 6s 29ms/step - loss: 0.0085 - accuracy: 0.99
86 - val_loss: 0.4533 - val_accuracy: 0.8610
Epoch 4/5
196/196 [==============================] - 5s 24ms/step - loss: 0.0011 - accuracy: 1.00
00 - val_loss: 0.4806 - val_accuracy: 0.8654
Epoch 5/5
196/196 [==============================] - 5s 24ms/step - loss: 4.6788e-04 - accuracy:
1.0000 - val_loss: 0.5020 - val_accuracy: 0.8661
```

In [11]:
```
pd.DataFrame(history.history).plot(figsize=(10,7))
plt.title("Model Metrics")
plt.show()
```



In [12]:
```
loss, accuracy = model.evaluate(x_test, y_test)
print("Test Accuracy:", accuracy)
```

```
782/782 [==============================] - 2s 3ms/step - loss: 0.5020 - accuracy: 0.866
1
Test Accuracy: 0.8661199808120728
```

In [13]:
```
y_pred = model.predict(x_test)
```

```
782/782 [==============================] - 3s 3ms/step
```

In [16]:
```
y_pred
```

Out[16]:
```
array([[4.4813273e-03],
       [9.9999833e-01],
       [3.5806501e-01],
       ...,
       [7.6598744e-03],
       [7.7849359e-04],
       [9.3456751e-01]], dtype=float32)
```

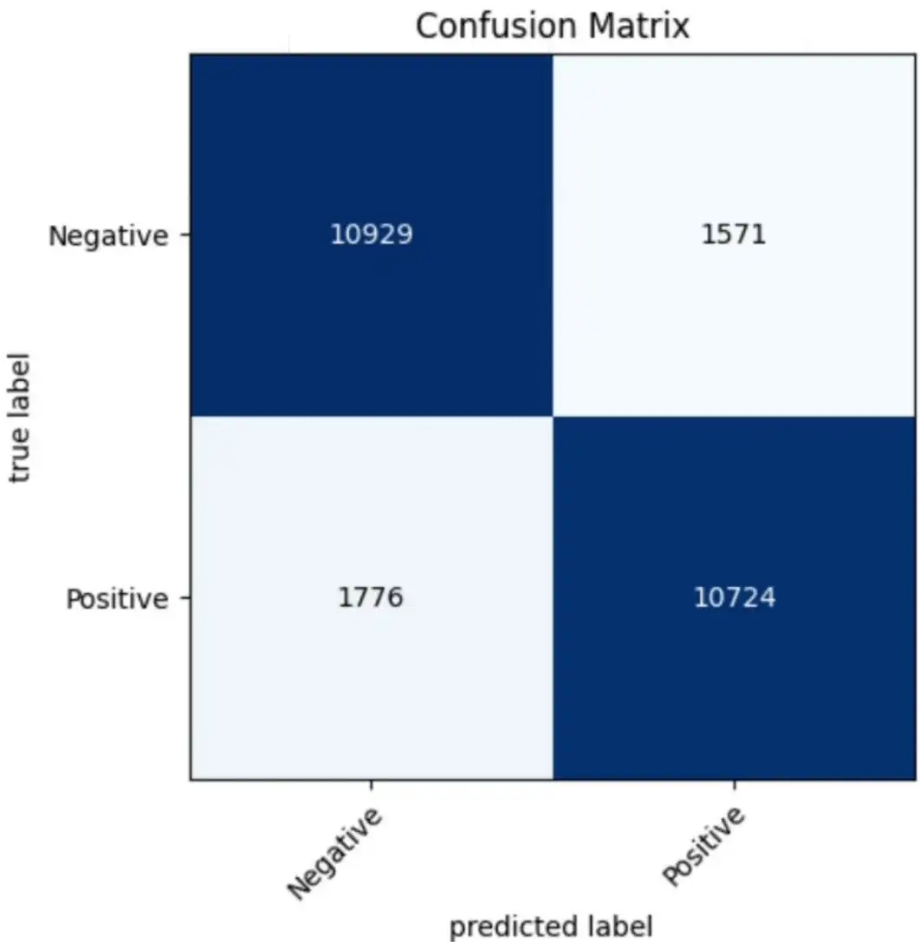
In [17]:
```
y_pred = y_pred.flatten()
```

In [19]:
```
y_pred
```

Out[19]:
```
array([4.4813273e-03, 9.9999833e-01, 3.5806501e-01, ..., 7.6598744e-03,
       7.7849359e-04, 9.3456751e-01], dtype=float32)
```

In [20]:
```
y_pred = (y_pred > 0.5).astype(int)
```

In [21]:
```
print(metrics.classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.86      0.87      0.87     12500
           1       0.87      0.86      0.87     12500

    accuracy                           0.87     25000
   macro avg       0.87      0.87      0.87     25000
weighted avg       0.87      0.87      0.87     25000
```

In [22]:
```
cm = metrics.confusion_matrix(y_test, y_pred)
plot_confusion_matrix(cm, class_names=['Negative', 'Positive'])
plt.title("Confusion Matrix")
plt.show()
```